VCT CONNECT

Revised MVP Technical Specification

AI-powered supplier-risk intelligence platform for Vietnamese sellers sourcing from China

Source product report: CP4 Final G4.pdf
Architecture decisions incorporated through 27 September 2026

Executive Summary

VCT Connect is an AI-powered supplier-risk intelligence platform for Vietnamese sellers and SMEs sourcing from 1688, Taobao, and Alibaba. The MVP converts a supplier or product URL into an evidence-backed Vietnamese report that highlights sourcing risk, review patterns, factory-versus-trading-company likelihood, and recommended pre-order actions.

This revision incorporates the product requirements from CP4 Final G4.pdf and the technical architecture decisions made after the original implementation plan. Source-derived requirements and engineering decisions are kept distinct so the team can see what comes from the business report and what has been chosen for implementation.

Core MVP question

Before I order from this Chinese supplier, what risks should I know about?

Source / engineering boundary: *Project concept pp. 5-13 and MVP plan pp. 42-45 define the product behavior. The concrete technology stack, scoring formula, authentication, queueing, AI gateway, and trial strategy below are engineering decisions added after the report.*

1. Implementation Goal

- Reduce the effort and uncertainty involved in checking Chinese suppliers before a buyer sends money or places a bulk order.
- Validate contextual Chinese-to-Vietnamese interpretation for B2B sourcing information.
- Validate review-risk analysis, including recurring complaints and suspicious review patterns.
- Validate factory-versus-trading-company assessment using probabilities, evidence, and confidence.
- Return a useful report in approximately 20-30 seconds for supported links whenever extraction conditions allow.
- Validate repeat usage, Watchlist behavior, extension adoption, and willingness to pay before enabling real-money billing.

Implementation adjustment: *The source report originally includes willingness-to-pay and free-to-paid conversion as MVP hypotheses. Because payment implementation is intentionally deferred, the trial will measure upgrade intent and usage first; actual paid conversion is measured after billing launches.*

Source basis: MVP objectives and success criteria, CP4 Final G4.pdf pp. 42-44.

2. MVP Scope and Trial Strategy

2.1 User-facing products

- VCT Connect Web Application.
- Chrome Extension connected to the same FastAPI backend and Clerk identity.

2.2 Initial source platforms

- 1688
- Taobao
- Alibaba

2.3 MVP features

- Supplier/product URL extraction and normalization.
- Vietnamese contextual summary and risk report.
- Review summarization, complaint extraction, suspicious-pattern detection, and multimodal review handling where accessible.
- Factory-versus-trading-company likelihood with evidence and confidence.
- Deterministic supplier-risk engine with separate Risk, Confidence, and Data Coverage outputs.
- Basic supplier Watchlist and analysis history.
- Internal Human Evaluation Console for model QA and dataset creation.
- Guest, registered-web, and extension-enhanced analysis modes.

2.4 MVP trial and monetization treatment

- No real-money payment gateway is required for the MVP validation period.
- The first intended paid tier is temporarily available free during the trial.
- Eligible trial participants receive one additional month of Premium when paid plans officially launch.

- No user is automatically charged at the end of the trial. Trial entitlements expire and users fall back to Free unless they explicitly purchase later.
- The exact definition of an eligible trial participant and the payment provider remain open decisions.

2.5 Explicitly deferred

- Real-money payment gateway and recurring billing.
- Custom-trained models.
- Direct logistics-company API integrations.
- Full Vietnamese Buyer Intelligence Database.
- Category-specific risk models.
- Advanced enterprise/team workflows.
- Continuous supplier monitoring and alerts.
- Tmall and Pinduoduo support.
- Large-scale premium human-review operations.

Source basis: MVP scope, pricing model, and augmented product, pp. 11-13, 32, 43. Trial mechanics are engineering/product decisions.

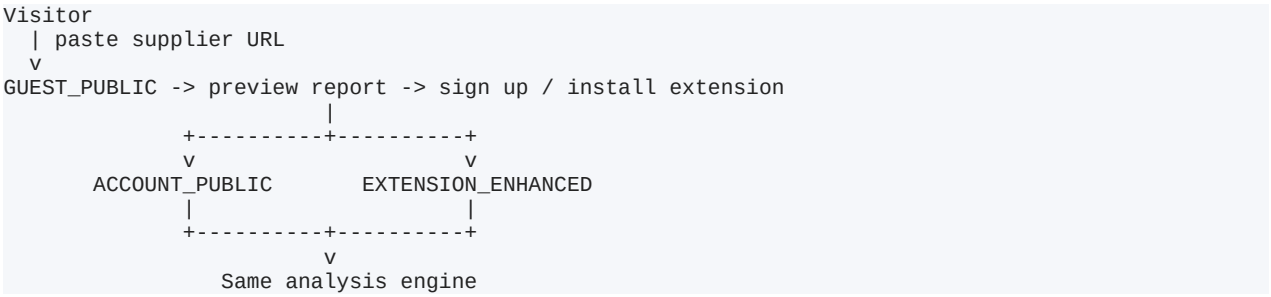
3. Locked Technical Decision Register

#	Decision	Locked choice / status
1	Frontend	Next.js + React + TypeScript for Web; React + TypeScript for Chrome Extension.
2	Backend	Python + FastAPI.
3	Database	PostgreSQL + JSONB.
4	Cloud	Microsoft Azure.
5	Extraction	Hybrid Guest / Registered / Extension model; httpx + selectolax; Playwright + Chromium fallback; extension-assisted extraction.
6	Jobs / queue	Azure Service Bus Standard + direct Python Azure SDK.
7A	AI gateway	YEScale behind an internal LLMProvider abstraction.
7B	Exact LLM(s)	Open for benchmarking using human-labeled evaluation data.
8	Custom models	Deferred until enough reliable labels and outcomes exist.
9	Authentication	Clerk; FastAPI verifies JWTs; business state stays in PostgreSQL.
10	Payment	Deferred until after MVP validation.
10A	Trial	First paid tier free during trial; eligible participants receive 1 month Premium at paid launch.
11	Risk engine	Deterministic evidence-based multi-dimensional scoring; versioned.
12	Factory/review methodology	Heuristics + embeddings + structured LLM interpretation; confidence/provenance; no custom model for MVP.
12B	Internal evaluation	Blind human evaluation of sampled text/image/video reviews; model/prompt/version metrics retained.

4. Product Modes and Customer Journey

The Web App and Extension are not merely two interfaces to the same crawl. They form a progressive evidence funnel with three analysis modes.

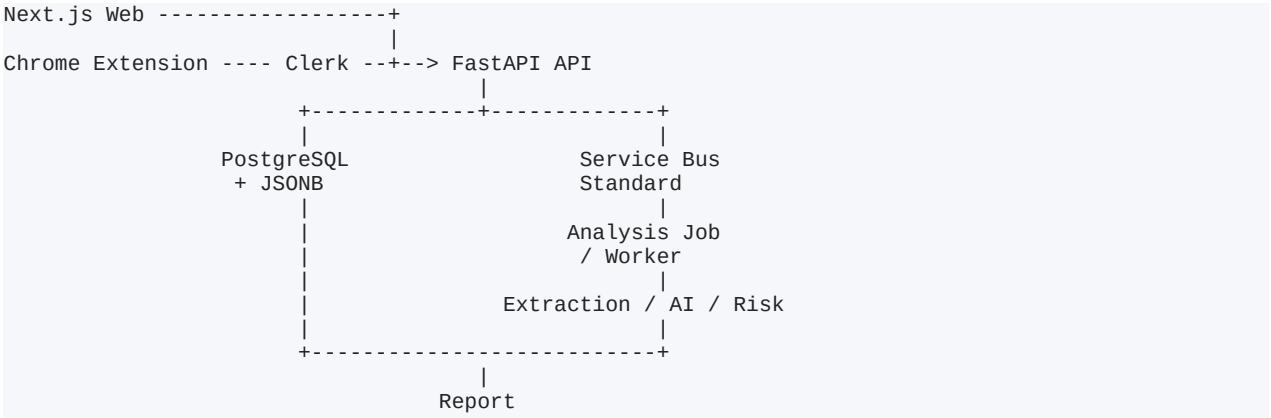
Mode	Authentication	Primary extraction	Product depth
GUEST_PUBLIC	No VCT account; no platform credentials provided	Server-side HTTP, then Playwright when justified	Limited public-data preview; show coverage/confidence; prompt to register/install extension.
ACCOUNT_PUBLIC	Clerk-authenticated VCT user	Same public server extraction	Fuller interpretation, history, Watchlist, trial/Premium entitlements.
EXTENSION_ENHANCED	Clerk sign-in required	Browser-rendered DOM/page state + server enrichment	Richest evidence available to VCT; full report and saved workflow.



The extension must not forward the user's 1688/Taobao/Alibaba password, cookies, or session tokens to VCT. It reads permitted rendered supplier/product/review data locally and sends only the data required for analysis. Server-side enrichment can then add public and historical VCT data.

Engineering decision: *Missing data never means a supplier is safe. Guest reports must visibly communicate reduced data coverage and lower confidence when extraction is limited.*

5. High-Level System Architecture



Supporting Azure services: Container Apps, Container Apps Jobs, Blob Storage, Key Vault, logging/monitoring.

Start as a modular monolith plus asynchronous analysis worker/job. Do not split into microservices until scale or ownership boundaries justify it.

6. Frontend Implementation

6.1 Web application

- Next.js + React + TypeScript.
- Public landing and guest URL submission.
- Authenticated dashboard, history, Watchlist, reports, account/trial status, and internal reviewer routes for authorized staff.
- Shareable/SEO-friendly report pages where appropriate.
- Use shared TypeScript domain types and API client with the extension.

6.2 Chrome Extension

- React + TypeScript; an extension framework such as WXT may be used.
- Detect supported pages and extract permitted DOM/page-state data.
- Clerk-authenticated; use web-to-extension session synchronization where practical.
- Show analysis progress and condensed verdict; open full report in the Web App.
- Remain a thin client: scoring, plan authorization, and sensitive logic stay in FastAPI/backend services.

```
frontend/
  web/
  extension/
  packages/
    ui/
    types/
    api-client/
```

7. Backend and Asynchronous Analysis

Use Python + FastAPI for REST APIs and keep extraction, AI/NLP, evidence logic, and risk scoring in the Python ecosystem.

A 20-30 second supplier analysis must not run as one synchronous HTTP request. The API creates an Analysis record and queues work.

```
POST /api/v1/analyses
  -> validate URL / entitlement
  -> create Analysis(status=QUEUED)
  -> send analysis_id to Service Bus
  -> return {analysis_id, status}

Azure Service Bus Standard
  -> Container Apps Job / worker
  -> extract -> normalize -> AI -> risk -> report
  -> persist result
  -> complete message

GET /api/v1/analyses/{id}
  -> frontend polls until COMPLETED / FAILED
```

- Use analysis_id as the Service Bus MessageId.
- Use Peek-Lock processing and complete the message only after the report is safely persisted.
- Workers must be idempotent; a repeated message must not create duplicate reports, duplicate entitlements, or duplicate AI spend.
- Repeated failures go to the built-in dead-letter queue for inspection/reprocessing.
- Do not introduce Celery for the MVP. Add task orchestration only if the pipeline later becomes a genuinely distributed workflow.

8. Azure Deployment

Component	Azure service	MVP role
Next.js Web	Azure Container Apps	Public and authenticated web UI.
FastAPI	Azure Container Apps	REST API, auth verification, entitlement checks, report retrieval.
Analysis workers	Azure Container Apps Jobs / event-driven worker	Consume Service Bus jobs and run extraction/AI/risk pipeline.
Queue	Azure Service Bus Standard	Durable analysis jobs, DLQ, duplicate detection/future messaging capabilities.
Database	Azure Database for PostgreSQL Flexible Server	Relational business data + JSONB snapshots/evidence.
Media/reports	Azure Blob Storage	Review images/videos and exported assets where retention is permitted.
Secrets	Azure Key Vault	YEScale, Clerk/backend, platform/service credentials.

Keep core services in one Southeast Asia region where possible to reduce latency and cross-region traffic. Exact SKU sizes and production cost assumptions should be verified at deployment because cloud pricing changes over time.

9. Extraction and Crawling Strategy

9.1 Hybrid extraction hierarchy

```
Supported URL
   |
   +--> Extension available? -> read permitted DOM/page state
   |
   +--> Server Web flow -> httpx + selectolax
                              |
                        enough data?
                          /      \
                        yes       no
                         |         |
                         |     Playwright + Chromium
                         +---------+
                                   |
                             Normalizer
                                   |
                             SupplierData
```

- Use one platform adapter per 1688/Taobao/Alibaba source.

- Prefer lightweight HTTP extraction for cost and latency; launch Chromium only when necessary.
- If a site requires authentication, presents a CAPTCHA, or explicitly blocks the server route, return AUTH_REQUIRED/BLOCKED and guide the user to the extension rather than building a CAPTCHA-bypass arms race.
- Do not send user platform cookies or session tokens to the backend.
- Store extraction method, extractor version, missing fields, coverage, and timestamps.

9.2 Extraction status model

```
SUCCESS | PARTIAL | AUTH_REQUIRED | BLOCKED | UNSUPPORTED_PAGE | TIMEOUT | PARSE_FAILED
```

9.3 Raw and normalized snapshots

```
SupplierSnapshot
  raw_payload        JSONB
  normalized_data    JSONB / relational refs
  extraction_method
  analysis_mode
  extractor_version
  completeness
  missing_fields[]
  extracted_at
```

Preserving raw snapshots allows later re-normalization if a platform parser was wrong without necessarily crawling the page again.

Source basis: supplier fields and extraction reliability requirements, pp. 7, 33-35, 44.

10. Normalized Supplier Data Contract

```
SupplierData
  supplier_name
  platform
  source_url
  company_information
  years_active
  categories[]
  certifications[]
  products[]
  price_information
  transaction_signals
  rating
  reviews[]
  delivery_information
  activity_history
  extraction_timestamp
  completeness
  missing_fields[]
```

The analysis pipeline must consume this common contract rather than platform-specific HTML. Platform-specific data that does not fit the common model can remain in JSONB for later use.

11. AI / LLM Architecture

11.1 Gateway decision

- YEScale is the locked AI gateway/control plane.
- All calls originate from backend workers; never expose the YEScale production key to the Web App or extension.
- Wrap access behind an internal LLMProvider interface so the rest of VCT is not coupled to YEScale-specific code.
- Store secrets in Azure Key Vault.
- Attach anonymous analysis/feature metadata for cost attribution where supported; do not send unnecessary personal identifiers.

```
Analysis modules
   |
   v
LLMProvider interface
   |
YEScale adapter
   |
   +--> candidate model A
   +--> candidate model B
   +--> fallback provider/model
```

11.2 Exact model selection remains open

Decision 7B is intentionally open. Candidate models will be benchmarked on Chinese-to-Vietnamese interpretation, complaint extraction, suspicious-review reasoning, factory/trader evidence extraction, multimodal review handling, structured-output reliability, latency, and cost per supplier report.

For core analytical tasks, pin a known candidate model/version during evaluation rather than blindly using changing automatic routing. YEScale may still handle controlled provider failover for availability.

11.3 Custom models

No custom-trained model is required for the MVP. Use third-party LLMs, deterministic rules, and off-the-shelf embeddings. Future custom-model candidates are factory/trader classification, suspicious-review anomaly detection, and risk calibration once VCT has reliable labels and buyer outcomes.

12. Review and Complaint Analysis

The system should identify recurring issues such as poor material quality, wrong size/color, missing quantities, delayed delivery, image mismatch, weak packaging, inconsistent batches, and poor refund/return support. It should report suspicious review patterns rather than claim that individual reviews are definitively fake.

12.1 Three-layer suspicious-pattern engine

Layer	Purpose	Examples
Deterministic statistics	Cheap, reproducible anomaly signals	Five-star bursts, timing concentration, exact duplicates, rating/text mismatch.
Off-the-shelf embeddings	Semantic similarity and clustering	Near-duplicate/template language, repeated complaint grouping.
Structured LLM interpretation	Language/context and multimodal interpretation	Complaint category, sentiment, severity, text-media consistency.

Review manipulation risk also reduces the confidence of review-derived Quality, Delivery, and After-sales evidence. A high platform rating should not be trusted strongly when the underlying review set itself appears manipulated.

```
ReviewAnalysis
  total_reviews
  accessible_reviews
  complaint_topics[]
  suspicious_patterns[]
  review_manipulation_risk
  review_reliability
  confidence
  evidence_ids[]
```

Source basis: Review and Complaint Analysis, pp. 8-9; transparency controls, p. 13.

13. Factory vs. Trading Company Methodology

Use a transparent evidence aggregator for the MVP rather than a custom classifier. Each signal stores direction (factory/trader/neutral), strength, reliability, and provenance.

Signal example	Direction	Typical strength	Reliability guidance
Registration includes manufacturing activity	Factory	Strong	High if independently verified.
Narrow, stable product specialization	Factory	Medium	Medium.
Broad unrelated catalog	Trader	Medium	Medium.
Verified manufacturing certificate	Factory	Strong	High.
Factory photos only	Factory	Weak	Low-Medium; easy to misrepresent.
Supplier self-claims "we are a factory"	Factory	Weak	Low; never ground truth by itself.

Output remains probabilistic: Factory likelihood, Trading-company likelihood, Confidence, Supporting evidence, and Uncertainty.

Future ground-truth labels should use verification tiers: VERIFIED_FACTORY, PROBABLE_FACTORY, INDETERMINATE, PROBABLE_TRADER, VERIFIED_TRADER, with a stored label source such as registration verification, platform verification, VCT manual review, logistics-partner verification, buyer report, or supplier self-claim.

Factory/trader status is not itself a supplier-risk penalty. A trader can be reliable and a factory can be poor. Only contradictory or misleading identity evidence should increase Supplier Identity Risk.

Source basis: Factory-versus-trading-company analysis, p. 8.

14. Supplier Risk Engine v0.1

14.1 Separate Risk, Confidence, and Coverage

Output	Meaning
Risk Score	How concerning the available evidence is.
Confidence Score	How strongly the available evidence supports the assessment.
Data Coverage	How much of the expected supplier evidence was actually obtained.

Missing evidence is UNKNOWN, never zero risk. Low-coverage analyses can therefore return INSUFFICIENT INFORMATION instead of appearing artificially safe.

14.2 Initial dimension weights

Risk dimension	Weight v0.1
Product Quality	25%
Supplier Identity	15%
Delivery	15%
Review Manipulation	15%
After-sales	15%
Pricing	10%
Communication	5%
Total	100%

These are heuristic starting values, not validated ground truth. They must be versioned and calibrated during the trial.

14.3 Calculation model

```
For dimension d:
R_d = sum(w_i * q_i * s_i) / sum(w_i * q_i)

Overall risk:
R = sum(W_d * C_d * R_d) / sum(W_d * C_d)

where:
s_i = signal severity (0-100)
q_i = evidence reliability
w_i = signal importance
W_d = dimension weight
C_d = confidence in usable evidence for that dimension
```

14.4 Confidence v0.1

Confidence factor	Initial contribution
Data coverage	40%
Source reliability	25%
Sample adequacy	20%
Data freshness	15%

14.5 User-facing thresholds

Condition	Display
0-34	LOW
35-64	MODERATE
65-100	HIGH
Coverage < 45% OR Confidence < 40%	INSUFFICIENT INFORMATION

14.6 Critical red flags

A high-confidence critical signal may impose a minimum risk floor so that a severe identity or other verified issue cannot be averaged away by benign dimensions. Critical overrides require strict evidence rules; one angry or ambiguous review must never trigger them by itself.

14.7 Versioning

```
v0.1.0 = MVP heuristic
v0.2.0 = early trial calibration
v0.x   = continued validation
v1.0   = validated production baseline
```

Every analysis stores scoring_version so old reports remain reproducible after weights or thresholds change.

Source basis: Supplier Risk Score pp. 7-8 and Transparency/Accuracy Controls p. 13. Numerical weights and thresholds are engineering decisions.

15. Evidence and Risk Report Design

Every material finding must retain evidence, source/provenance, confidence, and freshness. The product is a decision-support system, not a guarantee that a supplier is safe.

```
Report
  Supplier / Platform / Date
  Overall Risk + Confidence + Data Coverage
  Factory / Trader likelihood + confidence
  Key Risks
  Positive Signals
  Review Summary
  Vietnamese Supplier Summary
  Recommended Pre-order Actions
  Evidence / Sources
  Missing Information / Limitations
```

Recommended pre-order actions can include requesting a current factory/warehouse video, confirming material/color/size, defect replacement policy, lead time, packaging photos, manufacturer status, and a small test order before committing to volume.

Source basis: pre-order recommendations, shareable report, and transparency controls, pp. 9-13.

16. Internal Human Evaluation Console

VCT will include an internal interface for authorized team members to evaluate LLM verdicts on reviews containing text, images, and/or video. This is a core QA and dataset-building tool for the MVP, not a customer feature.

16.1 Blind evaluation workflow

```
Review evidence (text / image / video / rating / metadata)
      |
      v
Human reviewer labels independently
      | submit
      v
Reveal stored LLM verdict
      |
      v
Compare category / severity / evidence / confidence
      |
      v
Evaluation dataset + error analysis
```

Reviewers should not see the model verdict before submitting their initial judgment, except in a separate debugging mode, to reduce anchoring bias.

16.2 Annotation fields

- Sentiment: positive / neutral / negative / mixed.
- Complaint categories: material quality, wrong color, wrong size/specification, image mismatch, missing quantity, packaging, delivery, batch inconsistency, refund/after-sales, other.
- Severity: low / medium / high.
- Media evidence: supports / partially supports / contradicts / cannot determine.
- Suspicious-review indicators: duplicate wording, unnatural/formulaic language, rating-text mismatch, timing anomaly, none observed.

- Model assessment: correct / partially correct / incorrect / insufficient evidence.
- Human confidence: low / medium / high.
- Optional image finding notes and video start/end timestamps.

16.3 Evaluation sampling

Use a mix of random samples and targeted cases: low-confidence outputs, high-risk/high-confidence outputs, multimodal reviews, rule/LLM disagreements, and novel patterns. A subset should receive two independent human reviews; disagreements are adjudicated so VCT can distinguish model error from genuinely ambiguous tasks.

16.4 Model-run provenance

```
llm_review_run
  model_id / model_version
  prompt_version
  pipeline_version
  structured_schema_version
  temperature / settings
  output JSONB
  confidence
  latency_ms
  input_tokens / output_tokens
  cost
  created_at
```

This console becomes the primary benchmark mechanism for Decision 7B and a future source of labeled data for the custom models deferred in Decision 8.

17. Authentication, Authorization, and Entitlements

17.1 Clerk identity

- Clerk handles user identity for Next.js and the Chrome Extension.
- FastAPI verifies Clerk JWTs and maps the Clerk subject to a VCT User record.
- Guest analyses remain anonymous and do not create Clerk users automatically.
- The extension requires VCT authentication; use Clerk web/extension session synchronization where practical.

17.2 VCT business authorization

PostgreSQL, not Clerk metadata, is the source of truth for VCT plan, credits, roles, trial/Premium entitlements, Watchlists, reports, and usage. FastAPI authorizes every protected action.

```
Entitlement sources
  MVP_TRIAL
  MVP_REWARD
  PAID_SUBSCRIPTION   (future)
  PROMOTION           (future)
  ADMIN_GRANT
```

Internal roles should include at minimum CUSTOMER, INTERNAL_REVIEWER, and ADMIN. /internal/* APIs must enforce roles in FastAPI, not merely hide UI routes.

18. Payment Strategy - Deferred

No payment gateway is selected for the MVP. VNPay, payOS/VietQR, SePay, or another provider can be evaluated after the product has demonstrated repeat usage and willingness to pay.

The data model should nevertheless be compatible with a future signed-webhook/IPN payment flow: internal PaymentOrder -> dynamic checkout/QR -> provider confirmation -> verified webhook -> atomic entitlement/credit update. The browser return page must never be treated as proof of payment.

Trial commitment

During MVP validation, the first intended paid tier is free. Eligible trial participants receive 1 month of Premium at paid launch. Nobody is automatically charged when the trial or reward month ends.

19. Database Model

Entity	Key purpose / fields

User	id, clerk_user_id, email, role, created_at.
Entitlement	user_id, entitlement, source, starts_at, expires_at, status.
Supplier	platform, platform_supplier_id, name, source_url.
SupplierSnapshot	supplier_id, raw_payload JSONB, normalized_data JSONB, extraction_method, analysis_mode, completeness, extractor_version, extracted_at.
Review	supplier_id/snapshot_id, text, rating, date, metadata JSONB.
ReviewMedia	review_id, type IMAGE/VIDEO, blob/source reference, metadata.
Analysis	user_id nullable for guest, supplier_snapshot_id, mode, status, scoring_version, timestamps.
RiskFinding	analysis_id, risk_type, severity, confidence, explanation.
Evidence	finding_id, source, evidence payload/reference, reliability, freshness.
AnalysisScore	overall risk/confidence/coverage + seven dimensions + classification.
Report	analysis_id, generated_at, report data/export reference.
Watchlist	user_id, supplier_id, created_at.
LLMReviewRun	review_id, model/prompt/pipeline versions, output, tokens, latency, cost.
ReviewEvaluation	review_id, llm_run_id, reviewer_id, human labels, correctness, confidence, notes.
MediaAnnotation	evaluation_id, media_id, timestamps/region/finding.
PaymentOrder	Future-ready only; user, provider, product, amount, status, provider IDs.
CreditTransaction	Future ledger for purchased/consumed credits.

20. API Contract - MVP Surface

Method / path	Purpose
POST /api/v1/guest-analyses	Create limited anonymous public analysis.
POST /api/v1/analyses	Create authenticated web or extension analysis; checks entitlement/quota.
GET /api/v1/analyses/{id}	Return analysis status/progress/result reference.
GET /api/v1/reports/{id}	Fetch full authorized report.
GET /api/v1/me	VCT profile, role, plan/trial entitlements, usage.
GET/POST/DELETE /api/v1/watchlist	Manage saved suppliers.
GET /api/v1/history	Analysis history.
GET /api/v1/internal/evaluation/reviews/next	Reviewer: fetch next evaluation case.
POST /api/v1/internal/evaluation/reviews/{id}	Reviewer: submit human annotation.
GET /api/v1/internal/evaluation/stats	Internal evaluation metrics.

Polling every few seconds is adequate for MVP analysis progress. SSE/WebSockets can be added later if real-time UX justifies the extra complexity.

21. Security, Privacy, and Data Handling

- Never send 1688/Taobao/Alibaba passwords, browser cookies, or session tokens to VCT.
- Never expose Clerk or YEScale server secrets in frontend/extension bundles.
- Store secrets in Azure Key Vault.
- Minimize data sent to YEScale; use anonymous internal identifiers rather than real names/emails when attribution is needed.
- Use short-lived authenticated Blob access for internal review media rather than permanent public URLs.
- Define media retention and deletion rules before production and ensure they are compatible with source-platform terms and applicable privacy obligations.
- Rate-limit guest analysis and cap expensive Playwright/LLM usage to protect the MVP budget.
- Log evidence provenance and scoring version so important decisions are auditable.

22. Testing and Model Evaluation

22.1 Extraction tests

- Normal supported URL.
- Different page layouts.
- Missing supplier fields/reviews.
- Large review volume.
- Redirects and invalid URLs.
- Blocked/auth-required extraction.
- Timeout and changed HTML structure.
- Extension data merge and duplicate evidence handling.

22.2 AI / analysis validation

- Chinese-to-Vietnamese contextual interpretation quality.
- Complaint-topic precision/recall and severity agreement.
- Suspicious-review pattern agreement.
- Factory/trader evidence and probability consistency.
- Multimodal text-image-video agreement.
- Evidence accuracy and unsupported-claim rate.
- Risk-engine reproducibility for a fixed scoring version.
- High-confidence model errors, not only low-confidence cases.

22.3 Human evaluation loop

During the trial, regularly sample analyses/reviews for internal review. Track model agreement by model, prompt version, platform, complaint category, severity, and media type. Use the same labeled set to benchmark Decision 7B candidates.

23. Observability and Cost Controls

- Log extraction method, duration, completeness, missing fields, parse errors, block/login rate, and extractor version.
- Log analysis queue time, worker duration, Service Bus retry/dead-letter counts, and end-to-end latency.
- Log LLM model, feature, analysis_id, token usage, latency, and estimated/provider-reported cost through the YEScale layer where possible.
- Measure cost per completed supplier report and cost by feature (translation, review analysis, factory/trader, report generation).
- Avoid retry amplification: keep application retries conservative when YEScale/provider fallback already retries upstream calls.
- Treat AI and scraping/proxy/browser costs as likely larger variable costs than the queue itself.

24. MVP Performance and Validation Requirements

Metric	Source target / MVP treatment
Registered users	>= 100.
URLs analyzed	500-1,000 during MVP validation.
Report generation	Approximately 20-30 seconds for supported links under normal conditions.
Returning users	>= 30% source target.
Free-to-paid conversion	Source target >= 15%, but real payment is deferred; measure upgrade intent during trial and paid conversion after billing launch.
Extraction quality	Track success/partial/auth-required/blocked/parse-failed rates by platform.
Model quality	Track human agreement, high-confidence error rate, and multimodal agreement.

Primary technical acceptance criterion: a supported supplier/product link should reliably produce a useful, evidence-backed report in approximately 30 seconds, or transparently explain when evidence is insufficient.

Source basis: MVP success criteria, p. 44.

25. Recommended Development Sequence

1. Foundation: monorepo/repositories, shared types, Azure dev environment, PostgreSQL schema, Clerk, FastAPI skeleton, Service Bus connectivity.
2. Extraction: normalized SupplierData contract, 1688 adapter first, then Taobao/Alibaba; HTTP path, Playwright fallback, extension-assisted payloads, snapshot persistence.
3. Analysis: contextual translation, review complaint extraction, suspicious-pattern rules/embeddings, factory/trader evidence engine, Risk Engine v0.1.
4. Reporting: evidence-backed Vietnamese report, confidence/coverage display, recommended actions, Guest preview versus full report.
5. Product workflows: history, Watchlist, trial entitlements, extension sign-in/session sync, extension enhanced analysis.
6. Internal evaluation: reviewer roles, multimodal review console, blind annotation workflow, model provenance, evaluation metrics.

7. Pilot validation: recruit users, process target volume, observe cost/latency, manually review errors, collect buyer outcomes and willingness-to-pay evidence.

26. First-Month Implementation Sprint

Week	Primary focus
Week 1	Finalize schema and environments; FastAPI/PostgreSQL/Clerk/Service Bus baseline; 1688 HTTP extraction; snapshot persistence; failure states.
Week 2	Playwright fallback; Taobao/Alibaba basics; YEScale adapter; translation/review modules; factory/trader evidence; Risk Engine v0.1.
Week 3	Report UI; Guest/Account/Extension modes; Watchlist/history; trial entitlements; extension extraction and auth.
Week 4	Internal evaluation console; deploy pilot; process 500-1,000 URLs; measure latency/cost; review errors; user interviews; scoring/prompt calibration.

This is a sequencing target, not a fixed commitment; extraction difficulty on 1688/Taobao may require more time than a nominal four-week plan.

27. Post-MVP Roadmap

- Benchmark and lock the production LLM configuration (Decision 7B).
- Select and integrate a payment provider only after product validation.
- Activate paid plans; grant eligible trial participants their 1-month Premium reward.
- Build verified Vietnamese Buyer Intelligence data from feedback and actual order outcomes.
- Introduce custom classifiers only when reliable labels justify them.
- Calibrate risk weights statistically using outcome data.
- Add category-specific risk profiles, richer Watchlist/monitoring, partner integrations, team/enterprise workflows, and later platform expansion.

Source basis: one-year roadmap and augmented product, pp. 10-13, 45.

28. Open / Deferred Decisions

Decision	Current status	Trigger to decide
7B Exact LLM/model configuration	Open	Benchmark with internal human-labeled review/supplier cases; compare quality, structured-output reliability, latency, and cost.
8 Custom models	Deferred	Sufficient verified labels and buyer outcomes; clear quality/cost benefit over rules/LLMs.
10 Payment gateway	Deferred	MVP demonstrates repeat usage and willingness to pay; compare Vietnamese QR/gateway onboarding, fees, webhook reliability, recurring needs.
Trial eligibility threshold for 1-month reward	Open detail	Define before paid launch based on meaningful participation criteria.
Exact production Azure SKUs / scaling limits	Deployment detail	Set after load testing and real cost observations.

Appendix A. Analysis State Machine

```
CREATED
  -> QUEUED
  -> EXTRACTING
  -> ANALYZING
  -> SCORING
  -> REPORTING
  -> COMPLETED

Failure branches:
  PARTIAL_DATA  (analysis may continue with lower confidence)
  AUTH_REQUIRED / BLOCKED / UNSUPPORTED_PAGE
  FAILED_RETRYABLE -> Service Bus retry
  FAILED_FINAL -> DLQ / user-visible failure
```

Appendix B. Repository Sketch

```
vct-connect/
  frontend/
    web/
    extension/
    packages/
      ui/
      types/
      api-client/

  backend/
    app/
      api/
      models/
      schemas/
      services/
      extractors/
        http/
        browser/
        platforms/
      analysis/
        translation/
        review_risk/
        factory_classifier/
        risk_scoring/
      ai/
        provider.py
        yescale.py
      workers/
      db/
    tests/

  infra/
    azure/
  docs/
```

Source Document and Scope Note

CP4 Final G4.pdf - Experimental Entrepreneurship project report for VCT / VCT Connect. Product goals, user problems, risk categories, supplier fields, MVP targets, and roadmap items in this specification derive from that report. Technology choices and exact implementation mechanics are engineering decisions made during subsequent design discussions and are identified as such throughout this revision.